RAN Virtualization - Impact on Operator Economics, Ecosystem and Value Chain

Whitepaper

Abstract:

RAN virtualization is a trending topic in cellular communication market these days. This paper delves into the various RAN architectures, deployment scenarios, challenges, and the economic impact it has on the entire telecom ecosystem by rebalancing the power of the incumbents.

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Introduction

The annual revenue growth for mobile network operators is approximately 1% between 2018 and 2020. Meanwhile, mobile data traffic is expected to increase sevenfold between 2017 and 2022, growing at a 46% annual growth rate. 5G network capabilities are enabling ultra-low latency applications and massive machine type communications which will drive the volume and variety of data traffic higher. [1]

Projected growth in global mobile recenue

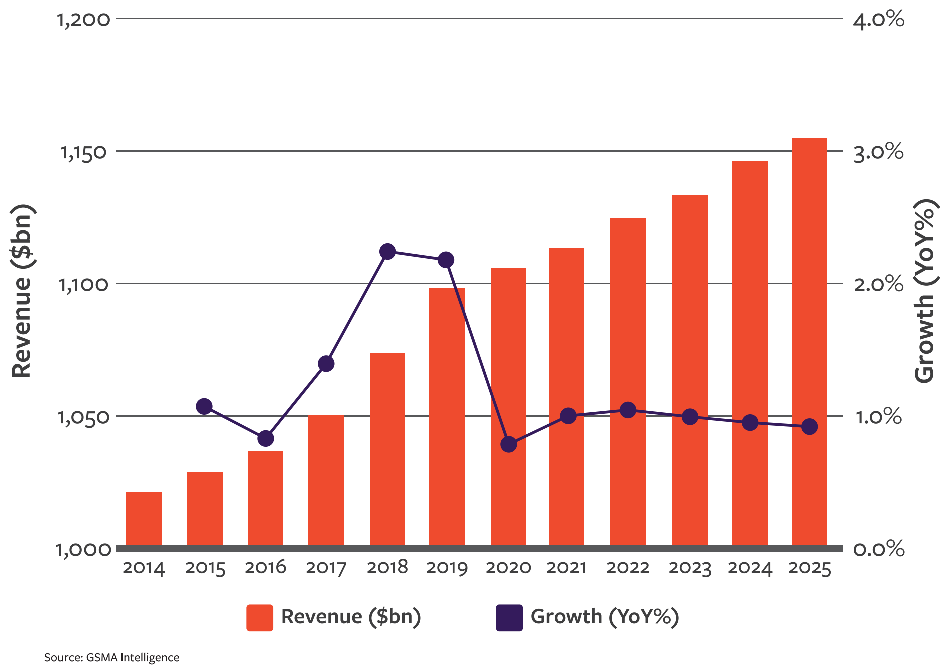

Figure 1. Mobile Revenue Growth and ARPU. Source: [1]

Mobile operators are under pressure to meet capacity demand while containing costs and launching new offerings in highly competitive mobile services markets. They now compete with internet giants like Amazon, Facebook and Google that offer similar, even more innovative, services at a far lower cost base and with agile development environments that speed time to market. [1]

80% of the MNO's Capital Expenditures (CAPEX) is spent on the RAN (distributed) to meet the rising demands of the growing number of users. Operating and maintaining costs (OPEX) of the RAN account for 60 % of the TCO. For MNO's, RAN is the most expensive part of the network, which is why it is very attractive to find a solution to reduce its TCO. [2]

With revenues expected to remain relatively flat, operators are forced to lower expenses. [3] Virtual RAN (vRAN) is giving the opportunity to operators to lower expenses and become something more than mere data pipes and enabling them to monetize opportunities through new services which were not possible hitherto. [4]

This paper deliberately uses the terminology **'vRAN'** as an umbrella term instead of Cloud-RAN. The Cloud-RAN is also a vRAN in which even the baseband is virtualized to a high degree. Between the legacy Distributed RAN (having proprietary hardware & applications) and a Cloud RAN there are many shades of virtual RANs. This paper addresses all of them to a certain degree.

RAN Architectures

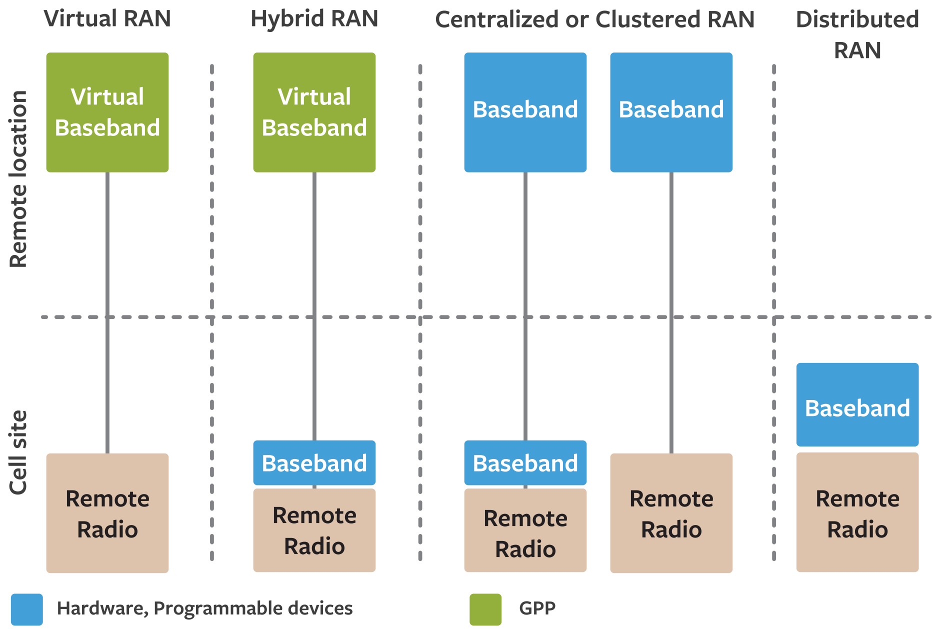

Figure 2. RAN Architectures [3]

Virtual RAN: The entire protocol stack runs in a remote location on general purpose processors and servers. Some implementations run the protocol stack on a processor without capabilities for pooling and load-sharing of resources (i.e. bare metal). [3]

Hybrid RAN: A split baseband architecture where some modem functions run on GPPs in the center while other baseband functions, such as Layer 1 or parts of Layer 2, run on programmable and hardware devices, such as FPGAs, DSPs, NPUs ASICs and SoCs, at the remote radio. The split can occur at different locations and is a vendor specific design. Hybrid RAN is an architecture that optimizes cost and performance but does not have the same disruptive potential as vRAN. [3] A detailed diagram is shown in Figure 3.

Clustered/Centralized RAN: All the base stations are located in a central location. This type of architecture in targeted for cost reduction. In some architecture, parts of the lower layers may be at the cell site while the higher layers are processed at a remote location (essentially split architecture without virtualization). [3]

Distributed RAN: Legacy RAN architecture. Protocol stack runs on proprietary hardware and each cell site houses a dedicated baseband unit. [3]

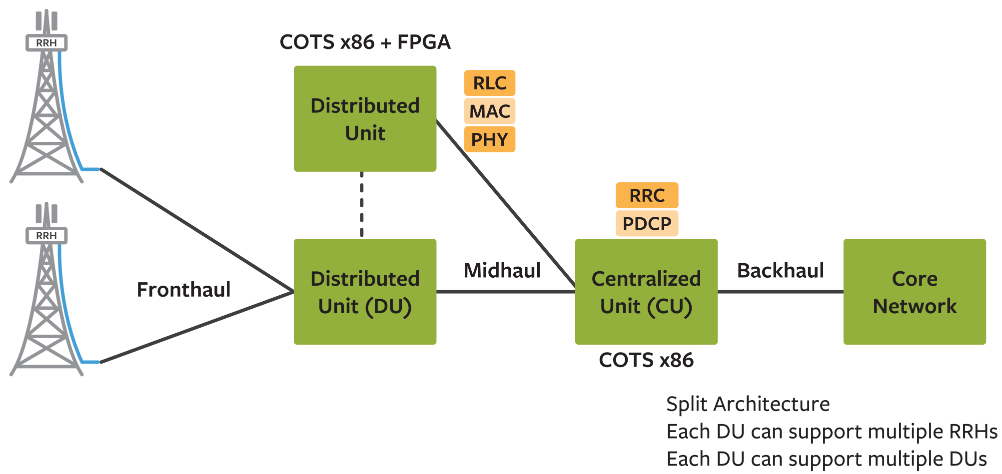

Figure 3. Hybrid RAN Architecture Source: [5][6]

RAN Virtualization Benefits

The vision for Network Functions Virtualization (NFV) was established by 13 of the world's largest telecom network operators in a white paper published in 2012. It proposed a radical transformation in the way networks are built and services are delivered in order to achieve [1]

a. Cost savings in Capex and Opex
b. Deployment flexibility
c. Service Agility/time to market.

Virtualization enables CSPs to deploy software-based network functions on general purpose hardware, rather than install proprietary appliances every time they need a new network function, service or application. [1] Virtual Radio Access Networks enables the baseband functions to be moved away from the cell site to a data center and allows pooling of baseband resources across multiple cell sites. This enables intelligent scaling of computing resources as demand on capacity fluctuates, while reducing costs. The benefits are listed below

1. Using COTS hardware for computing reduces Capex as proprietary equipment are costly
    a. In earlier architecture each cell site had to be built for peak capacity. In vRAN architecture resources can be pooled for multiple cell sites as all sites are not operating at peak capacity at any given time
    b. Moving to cloud completely enables MNOs to convert Capex into Opex
2. Baseband unit need not be kept at every antenna location. This reduces site leasing cost.
3. Enables intelligent scaling of computing resources as demand on capacity fluctuates
    a. This reduces deployment costs in case demand goes up
    b. Reduces energy consumption as operators don't have to build for peak capacity at every cell site
    c. Reduces maintenance costs
4. Easier (less time consuming) software upgrades
5. Cloud RAN is based on open platform, it opens the possibilities for new revenue sources with technologies such as Mobile Edge Computing (MEC) [2]

On top of cost savings, vRAN also brings performance benefits like coordinated multipoint (CoMP) and network MIMO. Base stations in Cloud RAN's BBU pool can work together and easily share signaling data, traffic data and Channel State Information (CSI) of active UE's in the network due to the nature of centralized baseband processing. Cloud RAN also enables easy implementation of joint processing and scheduling, which can help to mitigate inter-cell interference and improve spectral efficiency. The result is enhanced user experience, especially at the cell edge where performance is most lacking. [3] [2]

Value Chain Disruption

Virtualization decouples the software from hardware, enabling the use of commercial servers in the network. MNOs would no longer need to purchase hardware-optimized base stations from specific telecom equipment manufacturers (TEMs). Instead they would only need software and general-purpose servers in data centers to run the wireless protocol stack as an application to power any remote radios on demand. Other applications can run on the same infrastructure to provide value added services, such as video optimization, caching and localization. TEMs could provide their applications in a software as a service (SaaS) setting, with an OPEX-based pricing model, instead of the CAPEX-dominant model of today. MNOs could control and manage large networks more efficiently to enable a HetNet architecture. Because wireless capacity is not in demand at peak level at all locations at the same time, MNOs could save substantial expenses by multiplexing wireless capacity to increase operational efficiency and reduce capital costs. The RAN market structure will be radically changed, altering the balance of power between vendors and operators; leading new entrants into a market that's becoming highly consolidated. [3]

Some Case Studies

Senza Fili Consulting conducted a study (100 macro cells, 200 outdoor small cells & 250 indoor small cells). The study shows the vRAN implementation led to a weighted total cost of ownership (TCO) to reduce by 37% in deployment and operational costs over five years, derived from a 49% savings in capex in the first year, and an annual 31% savings in Opex over the full period in comparison to a distributed RAN. [1]

Compared to conventional, distributed RAN, Altiostar's vRAN software solution, which runs on industry standard x86 hardware and supports Ethernet fronthaul, reduces CAPEX by 40% to 60% and lowers OPEX by 30% to 40%. Compared to distributed RAN, vRAN reduces the total cost of ownership (TCO) over a seven-year period by up to 50%. [7]

Vodafone claims to have been able to reduce the cost to operate by more than 30 percent, using a much more open architecture, by being able to source components from different pieces when it tested the technology in India for 6 months. [8]

Solution/Deployment Scenario

A. Enterprise, Venue

Deployment of vRAN is likely to be driven by venues and indoor applications (including enterprises), where demand for capacity is highest. vRAN could be a substitute for small cells and DAS, which is not optimized to support MIMO technologies, a leading feature in LTE. [6] Currently, owners of large premises tend to invest in Distributed Antenna Systems (DAS) to deliver multi-operator coverage, which are expensive to deploy and upgrade. Smaller premises can provide indoor coverage with a femtocell, but that limits connectivity to one mobile operator. They are also not suitable for crowded public spaces such as shopping centers, stadia and hospitals. [9]

BT and Mavenir have developed a virtual radio access node that can reportedly host up to four operators, with each able to control and manage its own virtual segment of the cell as if it were its own dedicated infrastructure through its own media gateway. The solution is scalable and will be targeted at businesses and premises of all sizes. BT expects to offer the node as “a neutral, hosted solution, with an open fronthaul,” which will provide operators with more flexibility, and not require new sharing agreements between operators before any modification or update can be implemented. [9]

B. Private LTE

Virtualization has brought down the costs of deploying a private network in an Enterprise/Industrial environment. The opening of unlicensed bands (e.g. 5 GHz) and shared spectrum bands (e.g. 3.5 GHz CBRS and 2.3 GHz is also removing the barriers towards private LTE. A Private LTE network can support both human and machine communications on a single, reliable network that offers mobility without cumbersome portable radios and that opens up the world of the Internet of Things (IoT). [10]

C. Will vRAN displace Small Cells?

The Small Cell architecture is also intended to reduce cost per bit, by reducing the overall cost of baseband processing and radio hardware, as well as increasing spectral efficiency and overall throughput. Small Cells can drop the cost per bit by a factor of 4, compared to a macro LTE network. Even with high-cost backhaul such as millimeter-wave links, the cost per bit can drop in half. [11]
The Cloud RAN architecture is favored by operators with access to cheap fiber and in other cases only for stadium situations or other localized problems. The Small Cell architecture seems to be popular with most mobile operator CFOs, because the transport cost for RRH fiber outweighs the operational savings of co-locating the baseband processors. [11] Small cells will still be needed to cover dead zones or augment capacity in a specific zone. [12] So, small cells are not going to be displaced anytime soon.

Impact

A. Competition Intensification

Entry barriers for MNOs are gradually reducing with the availability of cheaper COTS hardware and some markets opening up the unlicensed bands for LTE. World over spectrum costs are very steep and one of the biggest expenses for an MNO. [13]

B. Telecom and Internet Ecosystem Convergence

Resolving the fronthaul cost challenge enables the Internet giants and fixed access service providers to enter the wireless market with lower cost basis, a move that is highly disruptive in a market dominated by telecom incumbents entrenched through massive equipment install-base. [3]

Behind the vRAN pioneers stand major Internet players such as Facebook, who initiated the Telecom Infrastructure Project (TIP) to explore the benefits of vRANs and its potential to reduce the cost of connectivity. TIP participants joined the Open Compute Platform (OCP) which is a 5-year old initiative on data center technologies for telecom companies. This points to the confluence of the Internet/compute world with the telecom world which has significant ramifications. [3]

C. Fixed Access Service Providers and Neutral Hosts

As LTE expands to unlicensed bands (e.g. 5 GHz) and shared spectrum bands (e.g. 3.5 GHz CBRS and 2.3 GHz), third parties will have the option to roll out LTE services there, concentrating on the indoor and venue markets. This allows companies with fixed assets such as fiber or cable, as well as neutral hosts, to enter the access service market with wireless solutions complementing those of the MNOs who own the wide-area coverage market. [3]

D. New Revenue Opportunities for MNOs - MEC (Multi-Access Edge Computing)

Edge presence is viewed as necessary to enable certain use case classes defined for 5G. The 5G use cases have been classified into three service types: [14]

- URLLC (Ultra Reliable and Low latency Communications): The low latency requirements prohibit the execution of URLLC use cases in the traditional “deep” or “remote” cloud. For example, V2X (Vehicle to anything) communication
- eMBB (enhanced Mobile Broad Band): eMBB service needs to support very high data rates; for example, high-definition video sharing, Augmented reality
- mMTC (Massive Machine Type Communication): mMTC set of use cases covers applications where a large number of IoT devices are sending data, collectively creating a significant data volume passing through the network. Moreover, this data is highly localized and is often associated with privacy requirements.

Therefore, all 5G use cases call for some processing of data and/or proximity at the edge of the Radio Access Network (RAN). [14]

It is not economically feasible to place huge servers at every base station. An MEC at the base station could just be used for some simple pre-computation to reduce the bandwidth required in the backbone network, with the actual processing being done deeper in the network. [15]

MEC will utilize the NFV infrastructure. MEC will also use (as much as possible) the NFV management and orchestration entities and interfaces. While NFV is focused on network functions, the MEC framework enables applications running at the edge of the network. Thus it will be beneficial for operators to reuse the infrastructure and infrastructure management of NFV to the largest extent possible, by hosting both VNFs (Virtual Network Functions) and MEC applications on the same platform. [16] Thus, operators can create a new revenue stream with the same set of investments.

Challenges

Capacity requirement for fronthaul: One of the biggest challenges of base station virtualization is meeting the strict demands of mobile signal real-time processing constraints in the virtual environment. This is because a typical BBU pool should support 10 - 1000 base stations. Transport of the I/Q samples from BBU to RRH requires a high fronthaul capacity. It requires 10x the capacity of an LTE backhaul channel, which makes it prohibitively expensive for operators who don't own fiber assets. CPRI also has tight requirements for synchronization, latency and jitter that are difficult to meet when there is no direct connectivity between baseband and radio. FDD LTE HARQ requires a round trip time (RTT) of 8ms that imposes an upper-bound for the sum of BBU processing time and the fronthaul transport latency. At a distance of 15 km, the overall processing time available will be between 2.3–2.6 ms (given that the speed of light in fiber is approximately 200 m/μs). [17]

As a result of these factors, fiber becomes the only media capable to implement fronthaul. While this is possible, especially as the cost and transmission capabilities of optical transceivers have been on a steep improvement curve, it remains a challenge to many operators who don't own fiber or where fiber penetration is thin. [3]
Altiostar claims to have implemented the fronthaul using Ethernet transport and tested it in several geographies with large operators. Fronthaul is supported on dark and lit fiber, FTTx, Ethernet, microwave and millimeter wave. [18] In Altiostar's vRAN solution, the eNodeB functional splits are designed to ensure low latency. In the higher layer split, real-time functions are integrated in the RRH, including the PHY, media access control (MAC) scheduler, and radio link control (RLC). Non-real-time functions are run on the centralized vBBUs, including packet data convergence protocol (PDCP), radio resource control and management, mobility management, IPSec, deep packet inspection, application intelligence, content caching and streaming, and analytics. In this way, the real-time functions are not hindered by latency conditions and bandwidth constraints in the fronthaul transport and CSPs have more options for transport infrastructure than just dark fiber, as is typically required in C-RAN deployments. [7]
Thus, it is possible to have a copper or ethernet fronthaul in a vRAN. But that would

mean some of the baseband functions have to be collocated with RRH. This partially beats the purpose of centralization and virtualization. However, this architecture may be used by operators who are constrained with fiber connectivity.

Virtualization: The wireless protocol stack includes computationally intensive functions that are inefficient to run on general purpose processors (GPPs). Devices such as FPGAs, ASICs and SoCs are more efficient, and provide real-time response capability, which is required by some RAN functions. Such challenges are beginning to dissipate as new, more powerful, GPPs with vector acceleration functions are becoming available on the market. Additionally, there are different implementations of virtualization that can solve these challenges such as offloading complex functions to acceleration engines. [3]

There are many reasons for NFV's as-yet unfulfilled promise: inadequate support for automation, complex integration with legacy systems, immature Management and Orchestration (MANO) systems, not enough incorporation of SDN programmability principles, just to name a few. The fundamental issue is that for the most part, the Virtual Network Functions (VNFs) themselves are not natively designed for high-performance cloud environments. Early VNFs were simply proprietary software that previously ran on customized appliances. The software wasn't rearchitected for virtualized environments. Today, most VNFs are built upon existing software code and modified to run in virtual machines (VMs) on general purpose hardware. But VMs have serious limitations in terms of scalability and resource utilization. [1]

In addition to above issues, C-RAN also brings many other challenges to BBU, RRH, and fronthaul. Front-haul multiplexing and topology, optimal mapping (clustering) between BBUs and RRHs, efficient BBU interconnections, cooperative radio resource management, energy optimization and harvesting techniques, and channel estimation are just few examples. [17]

Interoperability: Current telecom networks comprise multiple network functions from multiple vendors. Interoperability tests are required to verify VNFs work together nicely in a cloud environment. Even if different vendors say their products are based on standard platforms such as OpenStack, they might not necessarily interoperate well with each other. [19]

OpenRAN Radio Partner Ecosystem

Open interfaces are in the best interest of the overall industry. The formation of this ecosystem creates a challenger to the traditional RAN vendors. [6] Open interfaces enable multi-vendor deployments, enabling a more competitive and vibrant supplier ecosystem. Open interfaces are essential to enable smaller vendors and operators to introduce their own services or customize the network to suit their own unique needs. [20]

OpenRAN allows the deployment of open market (whitebox) remote radio unit (RRUs) to interwork with the virtualized cloud base band unit (vBBU) over Ethernet fronthaul (FH). With this implementation operators can break the stranglehold of closed proprietary specifications and the need to implement dark fiber for RRU front haul, which is a major entry barrier. They can continue to provide fronthaul and backhaul in the traditional ways like microwaves and IP/MPLS technologies. [6]

vRAN Ecosystem

Operators	AT&T, BT, KT, NTT Docomo, Softbank, Sprint, T-Mobile, Verizon
OSS/Core/SON/SDN	Amdocs, ComArch, HPE, Kapsch, KGPCo, Netscout
System Integrator	Amdocs, Kapsch, Tech Mahindra
Service Orchestrator	Amdocs, Ciena, HPE, Nokia
Radio Hardware	Altiostar, Bai Cells, Blue Danube, CommScope, MTI Mobile
Virtual Network Functions	ASOCS, Altiostar, Amarisoft, ASTRI, Dali, Ericsson, JMA Wireless, Mavenir, NEC, Nokia, Parallel Wireless, Phluido,
LTE Stack	Aricent, Radisys
Baseband Hardware	Artesyn, Advantech, DELL, HPE, Kontron, SuperMicro
System on Chip	ARM, Cavium, eASIC, Intel

Figure 4. VRAN Ecosystem [21][3][7]

Conclusion

D-RAN is not going away very soon. Operators have invested a lot of money and will not uproot the existing infrastructure overnight for a virtual RAN unless the business case is very clear. Besides, getting access to cheap fiber is also a challenge. Existing operators will incrementally move towards vRAN and some new MNOs may start greenfield network deployment with a virtual network because they don't have the baggage of legacy architecture.

vRAN (with baseband pooling) or Cloud RAN is economical when there is a high user density. vRAN in a sparsely populated rural area with low potential for growth does not make any sense. [2] That does not mean that the TEMs can continue to sell their legacy D-RAN equipment. The market will probably gravitate towards some virtualization wherein the real time functions are integrated with the RRH and the non-real time functions run on GPP which is collocated with the RRH and antenna.

Cloud RAN's cloud-based architecture enables radio networks to be opened for new services and applications through open Application Programming Interfaces (APIs) and access channels. This allows open innovation and collaboration with different businesses on the mobile edge, which will help MNOs to capture more value, differentiate and achieve a competitive edge over their competition. [2]

Abbreviations

API	Application Programming Interface
APRU	Average Revenue per User
BBU	Baseband Unit
CAGR	Cumulative Annual Growth Rate
CAPEX	Capital Expenditures
CoMP	Co-operative Multi-Point processing technology
COTS	Commercial off-the-shelf
C-plane	Control Plane
CPRI	Common Public Radio Interface
C-RAN	Centralized Radio Access Network
CSP	Communications Service Provider
D-RAN	Distributed Radio Access Network
eNB	Evolved NodeB
HARQ	Hybrid Automatic Repeat Request
ICIC	Inter-Cell Interference Coordination
IoT	Internet of Things
LTE	Long Term Evolution
M2M	Machine-to-Machine
MAC	Medium Access Channel
MEC	Mobile Edge Computing
MIMO	Multiple Input Multiple Output
MMTC	Massive Machine Type Communication
MNO	Mobile Network Operator
NFV	Network Function Virtualization
Opex	Operating Expense
TCO	Total Cost of Ownership
UE	User Equipment
URLLC	Ultra Reliable and Low Latency Communications
vRAN	Virtual Radio Access Network

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About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Smart Devices & Wearables, Enterprise Grade Devices, SatCom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of over a 100 Fortune 500 companies, powering over a billion devices through its services and IP.

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